

PK – Polyketone PK

AKROTEK® PK-HM black (4953)

Tensile modulus

1500 MPa

1 mm/min

ISO 527-2

Stress at yield

60 MPa

50 mm/min

ISO 527-2

Charpy impact strength

n.b. kJ/m²

23°C

ISO 179-1/1eU

AKROTEK® PK-HM black (4953) is an unreinforced Polyketon with average flowability.

Typical applications

Mainly components in mechanical engineering and the automotive industry.



Mechanical Properties

Tensile modulus (1 mm/min | ISO 527-2)

d.a.m.

1500 MPa

conditioned

1500 MPa

Stress at yield (50 mm/min | ISO 527-2)

d.a.m.

60 MPa

conditioned

60 MPa

Strain at break (50 mm/min | ISO 527-2)

d.a.m.

>200 %

conditioned

>200 %

Charpy impact strength (23°C | ISO 179-1/1eU)

d.a.m.

n.b. kJ/m²

conditioned

n.b. kJ/m²

Charpy notched impact strength (23°C | ISO 179-1/1eA)

d.a.m.

15 kJ/m²

conditioned

15 kJ/m²



Thermal Properties

Temperature of deflection under load HDT/A (1,8 MPa | ISO 75)

100 °C

Melting temperature (DSC, 10K/min | DIN EN 11357-1)

220 °C



Flammability

Burning rate (<100 mm/min) (> 1 mm Thickness | FMVSS 302)

+



General properties

Density (23°C | ISO 1183)

1,24 g/cm³

Humidity absorption (70°C, 62% r.F. | ISO 1110)

0,8-0,9 %

Molding shrinkage (flow | ISO 294-4)

1,4-1,6 %

Molding shrinkage (transverse | ISO 294-4)

1,5-1,7 %

Disclaimer:

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Electrical Properties

Surface resistivity (acc. to IEC 60093)
d.a.m.
conditioned

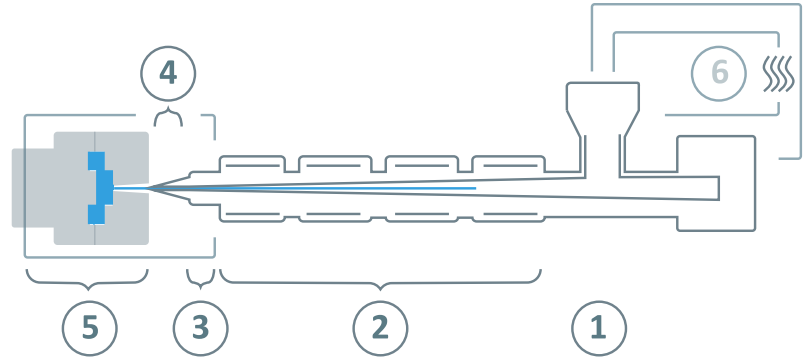
1E+13 Ohm
1E+10 Ohm

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Processing information

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only dehumidifying or vacuum dryers. Extensive drying can cause filling problems and surface defects.



⑥	Drying time	0 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80°C
	Processing moisture	0,02 - 0,1%
①	Feed section	60 - 80°C
②	Temperature zone 1 - Zone 4	220 - 250°C
③	Nozzle temperature	230 - 250°C
④	Melting temperature	230 - 250°C
⑤	Mold temperature	60 - 120°C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	30 - 70 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

Warning

The molding machine needs to be purged with polyolefines before and after processing of AKROTEK® PK! There is a risk of cross linking caused by reactions with POM or PA as well as unsuitable master batches! Cross linking is visible through dark spots! In this case purge immediately with polyolefines.

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